

# Hospital Management System - Project Report

## • Author Details

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About : I love developing applications that simplify everyday life for people.

## • Project Details

- **Project Title : Hospital Management System**
- **Project Statement :**  
HMS-V2 web application will allow efficient management of **appointments like booking, tracking, notifying and managing patients, doctors, and treatment records** in one unified place.
- **Approach :**  
I have used JWT for role based authentication and authorization in my Flask backend for efficiently identify and manage different users (admin, patient, doctor) resources. And Vue Js for dynamically handling different users' views in my frontend.

## • Frameworks and Technologies Used

### → Backend :

- **Flask** : Python microweb framework used as backend
- **Flask-SQLAlchemy** : Flask extension for Object Relational Mapper for SQLite database
- **Flask-JWT-Extended** : Flask extension to implement token based authentication and authorization
- **Flask-Mail** : Flask extension to send email from the backend server
- **Flask-Restful** : Flask extension to implement REST APIs
- **Flask-Cors** : Flask extension to implement CORS policy for the application
- **Flask-Caching** : Flask extension to send email from the backend server
- **Jinja2** : Templating engine for HTML pages (used for doctor report creation only)
- **Sqlite3** : File based serverless relational database
- **Redis** : The in-memory database for caching and as a message broker
- **Celery** : It's a task queue with focus on real-time processing, while also supporting task scheduling
- **Redis-py** : Python client library for redis database interactions
- **Bcrypt** : Python library for password hashing
- **Email\_validator** : Python library to validate emails in the backend

## → Frontend :

- **Vite** : Javascript project building and scaffolding tool
- **Vue3** : The Progressive JavaScript Framework for building web interfaces
- **Vue-Router** : Vue official extension for frontend application routing
- **Vue-Pinia** : Vue official extension for application state management
- **Vue3-Select-Component** : Vue3 extension for custom select input
- **Axios** : JavaScript library used to make HTTP requests
- **Bootstrap5** : CSS framework for styling and designing responsive UI
- **Bootstrap-Icons** : The collection of SVG icons package by bootstrap

## • **ER Diagrams**

### → Login and Registration Schema: ([ER-diagram\\_HMS-Login\\_Schema.png](#))

- ◆ **User** – Stores user credentials ( id, email, password, is\_active, fs\_uniquifier )
- ◆ **Role** – Stores the roles ( id, name, description )
- ◆ **Roles\_Users** – The joining table between role and user ( id, user\_id, role\_id )
- ◆ **User\_Tokens** – Stores all JWT generated for user ( id, user\_id, jti, type, create\_datetime, expiry\_datetime, valid )
- ◆ **Patient** – Stores the patient user data ( id, user\_id, public\_id, full\_name, contact, dob, gender, height\_cm, weight\_kg )
- ◆ **Doctor** – Stores the doctor user data ( id, user\_id, public\_id, full\_name, gender, license, qualification, experience, description, contact )

#### • **Relationships:**

- **One-to-One** : **User – Patient**
- **One-to-One** : **User – Doctor**
- **One-to-Many** : **User → User\_Tokens**
- **Many-to-Many** : **User → Roles\_Users ← Role**

### → Patient Schema: ([ER-diagram\\_HMS-Patient\\_Schema.png](#))

- ◆ **Appointment** – Stores the patient and doctors appointment schedules ( id, patient\_id, doctor\_id, date, start\_time, end\_time, status )
- ◆ **Treatment** – Stores the patient treatments for an appointment ( id, appointment\_id, visit\_type, tests\_done, diagnosis, prescription, medicine, notes )
- ◆ **Patient** – Stores the patient user data ( id, user\_id, public\_id, full\_name, contact, dob, gender, height\_cm, weight\_kg )
- ◆ **Doctor** – Stores the doctor user data ( id, user\_id, public\_id, full\_name, gender, license, qualification, experience, description, contact )

#### • **Relationships:**

- **One-to-One** : **Treatment – Appointment**
- **One-to-Many** : **Patient → Appointment**
- **One-to-Many** : **Doctor → Appointment**

→ **Doctor Schema:** ([ER-diagram\\_HMS-Doctor\\_Schema.png](#))

- ◆ **Availability** – Stores the doctor availability ( id, doctor\_id, slot\_id, date, start\_time, end\_time, diagnosis\_limit, appointment\_count, status)
- ◆ **Department** – Stores the departments ( id, name, description, type )
- ◆ **Departments\_Doctors** – The joining table between department and doctor ( id, doctor\_id, department\_id )
- ◆ **Specialization** – Stores all specializations ( id, name )
- ◆ **Doctor** – Stores the doctor user data ( id, user\_id, public\_id, full\_name, gender, license, qualification, experience, description, contact )

- **Relationships:**

- **One-to-Many** : **Doctor** → **Availability**
- **One-to-Many** : **Specialization** → **Doctor**
- **Many-to-Many** : **Doctor** → **Departments\_Doctors** ← **Department**

→ **Notification Schema:** ([ER-diagram\\_HMS-Notification\\_Schema.png](#))

- ◆ **Notification** – Stores the user notifications ( id, user\_id, date, time, data, type )
- ◆ **User** – Stores user credentials ( id, email, password, is\_active, fs\_uniquifier )

- **Relationships:**

- **One-to-Many** : **User** → **Notification**

- **API Resource Endpoints**

→ **Login And Registration APIs:**

|                            |      |                                 |
|----------------------------|------|---------------------------------|
| /api                       | GET  | API index                       |
| /api/register/patient      | POST | Registers Patient               |
| /api/register/doctor       | POST | Registers Doctor                |
| /api/login                 | POST | Login the User                  |
| /api/token/refresh         | GET  | Refreshes Tokens                |
| /api/token/user/role-valid | POST | Checks the token validity       |
| /api/logout                | POST | Logout the User                 |
| /api/logout/all            | POST | Logout the User from everywhere |

→ **Admin APIs:**

|                                   |            |                             |
|-----------------------------------|------------|-----------------------------|
| /api/dashboard/admin              | GET        | Gets the admin data         |
| /api/dashboard/admin/appointments | GET, PATCH | All appointments management |
| /api/dashboard/admin/patients     | GET        | Gets all the patient        |

|                                           |                          |                              |
|-------------------------------------------|--------------------------|------------------------------|
| /api/dashboard/admin/patient-appointments | GET                      | Gets the patient appointment |
| /api/dashboard/admin/patient-treatment    | GET                      | Gets the patient treatment   |
| /api/dashboard/admin/doctors              | GET                      | Gets all the doctor          |
| /api/dashboard/admin/stats                | GET                      | Gets the stats               |
| /api/dashboard/admin/specializations      | GET                      | Gets the specialization      |
| /api/dashboard/admin/doctor               | GET, POST, PATCH, DELETE | Manages the doctors          |
| /api/dashboard/admin/patient              | GET, POST, PATCH, DELETE | Manages the patients         |
| /api/dashboard/admin/notifications        | GET, DELETE              | Manages the notification     |
| /api/dashboard/admin/patient              | GET, POST, PATCH, DELETE | Manages the patients         |
| /api/dashboard/admin/appointments/search  | GET                      | Searches the appointments    |
| /api/dashboard/admin/patients/search      | GET                      | Searches the patients        |
| /api/dashboard/admin/doctors/search       | GET                      | Searches the doctors         |

→ Doctor APIs:

|                                                |                  |                             |
|------------------------------------------------|------------------|-----------------------------|
| /api/dashboard/doctor                          | GET              | Gets the doctor data        |
| /api/dashboard/doctor/availability             | GET, PATCH       | Manages doctor availability |
| /api/dashboard/doctor/appointments             | GET, PATCH       | Manages doctor appointments |
| /api/dashboard/doctor/week-stats               | GET              | Gets the week stats         |
| /api/dashboard/doctor/assigned-patients        | GET              | Gets the assigned patients  |
| /api/dashboard/doctor/patient-history          | GET, POST, PATCH | Manages the patient history |
| /api/dashboard/doctor/notifications            | GET, DELETE      | Manages the notifications   |
| /api/dashboard/doctor/assigned-patients/search | GET              | Searches the patient        |
| /api/dashboard/doctor/appointments/search      | GET              | Searches the appointments   |

→ Patient APIs:

|                                                     |             |                                    |
|-----------------------------------------------------|-------------|------------------------------------|
| /api/dashboard/patient                              | GET         | Gets the patient data              |
| /api/dashboard/patient/doctor-list                  | GET         | Gets the department doctors        |
| /api/dashboard/patient/upcoming-appointments        | GET, PATCH  | Manages upcoming appointments      |
| /api/dashboard/patient/departments                  | GET         | Gets the department                |
| /api/dashboard/patient/appointment-history          | GET         | Gets the patient appointments      |
| /api/dashboard/patient/appointment-treatment        | GET         | Gets the patient treatment         |
| /api/dashboard/patient/book-appointment             | GET, POST   | Manages the appointment booking    |
| /api/dashboard/patient/notifications                | GET, DELETE | Manages the notifications          |
| /api/dashboard/patient/upcoming-appointments/search | GET         | Searches the upcoming appointments |
| /api/dashboard/patient/departments/search           | GET         | Searches the departments           |
| /api/dashboard/patient/doctor-list/search           | GET         | Searches the doctor list           |
| /api/dashboard/patient/appointment-history/search   | GET         | Searches the appointment history   |
| /api/dashboard/patient/export-csv                   | GET, POST   | Requests for report export         |
| /api/dashboard/patient/download-csv                 | GET         | Downloads the CSV report           |

→ API documentation link - [HMS\\_V2\\_API\\_Documentation.yaml](#)

- **Presentation Video Link**

→ Google Drive link to the video : [🔥 HMS\\_V2\\_Video\\_Demo.mp4](#)